
# KOUSTAV MONDAL

**Address:** Ranaghat, West Bengal
**Phone:** +91 8391875038
**Email:** koustavmondal9641@gmail.com

**Portfolio:** dekoustav.vercel.app
**LinkedIn:** www.linkedin.com/in/koustav-mondal
**GitHub:** github.com/Kr4ken99z


## SUMMARY


Final-year Electronics & Communication Engineering student transitioning into Software Development. Strong foundation in Java, Data Structures & Algorithms, Spring Boot, MySQL, IoT, and Embedded Systems. Built full-stack and IoT-based projects with hands-on experience in backend development, ESP8266/ESP32, and real-world engineering applications. Seeking Software Engineer opportunities to build scalable solutions


## EDUCATION

**B.Tech in Electronics and Communication Engineering**
Academy Of Technology — **2022 - Present**

**Higher Secondary (80%)**
Purnanagar Purnachandra High School,Ranaghat — **2020 - 2022**

**Secondary (85%)**
Krishnagar Dharmachandra High School,Ranaghat — **2019 - 2020**

## SKILLS

**Languages:** Java, C, SQL, JavaScript
**Frontend:** HTML5, CSS3, JavaScript
**Backend:** Spring Boot, Spring Security, REST APIs
**Database:** MySQL
**Tools:** Git, GitHub, MATLAB, Simulink, Arduino IDE, ESP8266, ESP32, Excel
**Concepts:** OOP, Data Structures & Algorithms, Problem Solving

## WORK EXPERIENCE

**Industrial Trainee- Indian Railways(Signal & Telecommunication Dept) | Jul 2025**

- Observed and assisted in Electronic Interlocking (EI) and Route Relay Interlocking (RRI) operations.
- Studied Optical Fiber Communication infrastructure and railway signaling systems.
- Participated in maintenance activities, relay room inspections, cable laying, and control panel operations.

## PROJECTS

**AI-Powered Code Reviewer**
Tech Stack: Java, Spring Boot, React, MySQL, OpenAI/Gemini API, JWT, Docker

- Developed an AI-assisted platform that analyzes source code and provides automated review feedback.
- Implemented bug detection, code-quality scoring, optimization suggestions, and complexity analysis.
- Built secure authentication using JWT and role-based access control.
- Designed REST APIs and integrated AI models for real-time code evaluation.

**Intelligent Street Light for Smart City (Final Year Project)**
Tech Stack: ESP8266, ESP32-CAM, Arduino IDE, LDR, IR Sensors

- Developed an IoT-based automated street-lighting system for smart-city applications.
- Enables intelligent lighting control based on environmental and motion data.
- Designed to reduce power consumption and maintenance costs.

**Expense Tracker Application**
Tech Stack: Java, Spring Boot, Spring Security, MySQL, HTML, CSS, JavaScript

- Developed a secure expense management platform with authentication and database integration.
- Implemented income/expense tracking and CRUD operations.
- Designed backend APIs and persistent storage using MySQL.

## Language

English, Bengali, Hindi